

Application: Statistics in the Real World



Review

- Data Preparation
- Data Visualization
- Statistics Terminology and Key Ideas
- Hypothesis Testing
- Linear Regression

Statistical Terminology and Ideas

- Mean
- Median
- Mode
- Quantiles
- Range
- Frequency

Mode

The **mode** is the value that appears most often in a set of data.

The **range** is the difference between the lowest value and the highest value.

Range

Median

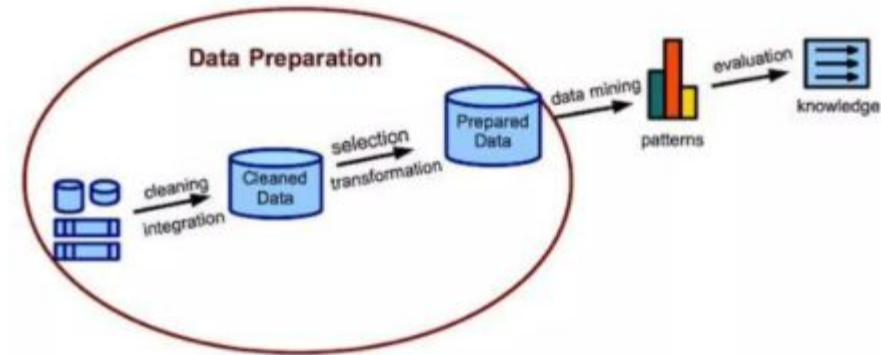
The **median** is the middle number in a list of numbers ordered from lowest to highest.

The **mean** is the total of all the values, divided by the number of values.

Mean

Data Preparation

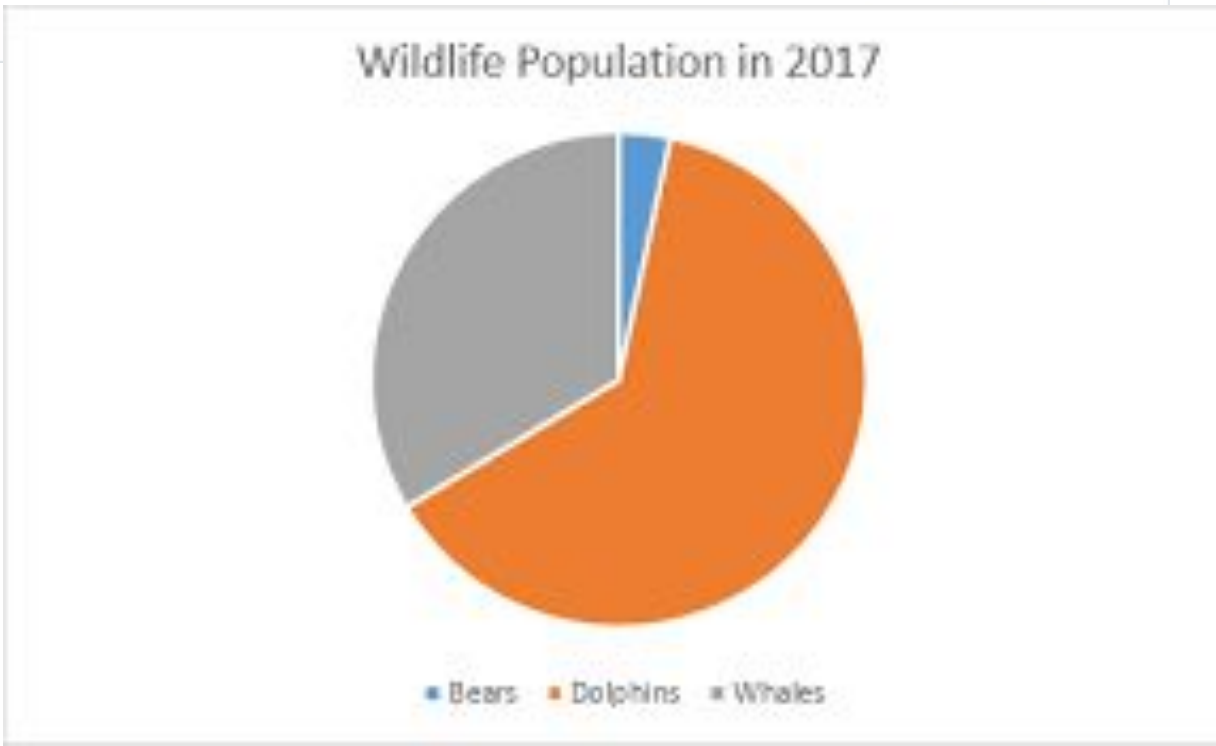
- Data Cleaning:
Removing NAs/Unknown values
- Removing unrealistic values
- Selecting certain columns to use
- Use of Binary variables



- There are multiple values that are commonly used to represent the virus. A virus like COVID-19 could be represented by 'SAR-Cov2', 'Corona', 'Covid' or 'Covid-19' to name a few.
- A data preparation tool could be used in this scenario to identify an incorrect number of unique values (in the case of virus, a unique count greater than suitable number in case of covid would raise a flag, as there are only few names aligned with virus). These values would then need to be standardized to use only an abbreviation or only full spelling in every row.

Data Visualization

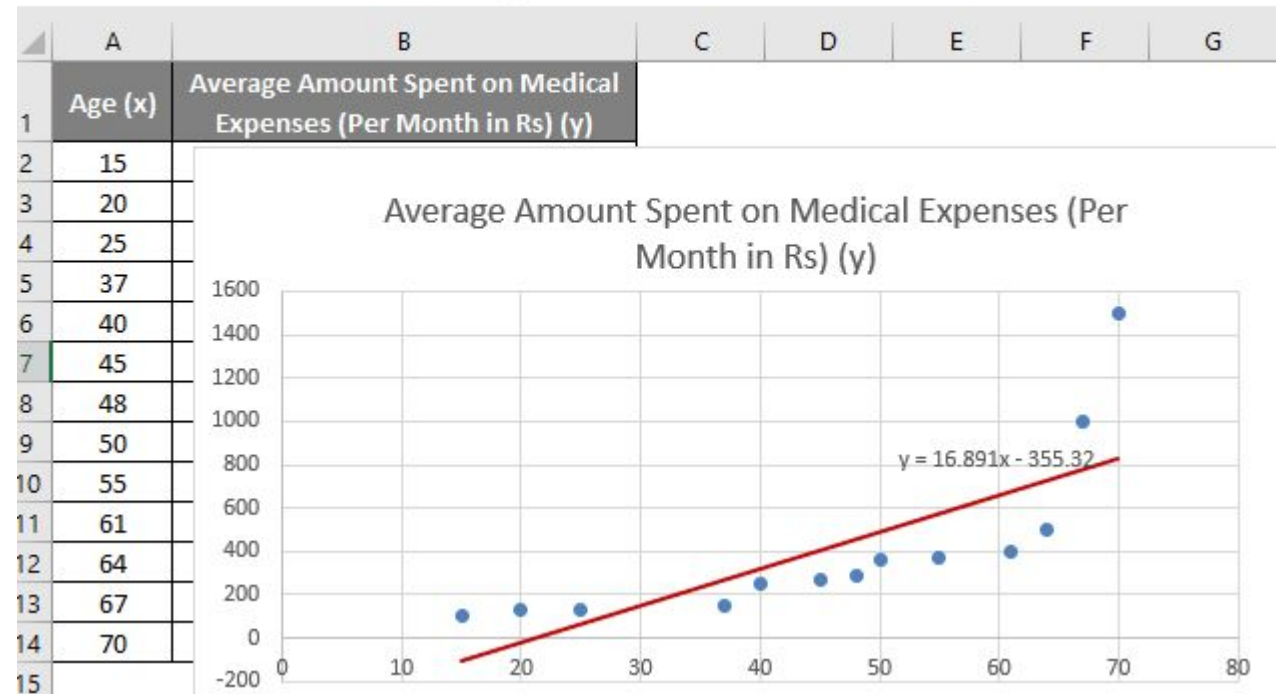
- Bar Graphs
- Line Graphs
- Pie Charts
- Axes Labels and Main Titles
- Legends



Linear Regression

- R-Squared Value
- Goodness of Fit
- Slope and Y-Intercept
- Positive v Negative Relationship
- Multiple Linear Regression: Multiple predictors for your response variable

Linear Regression in Excel

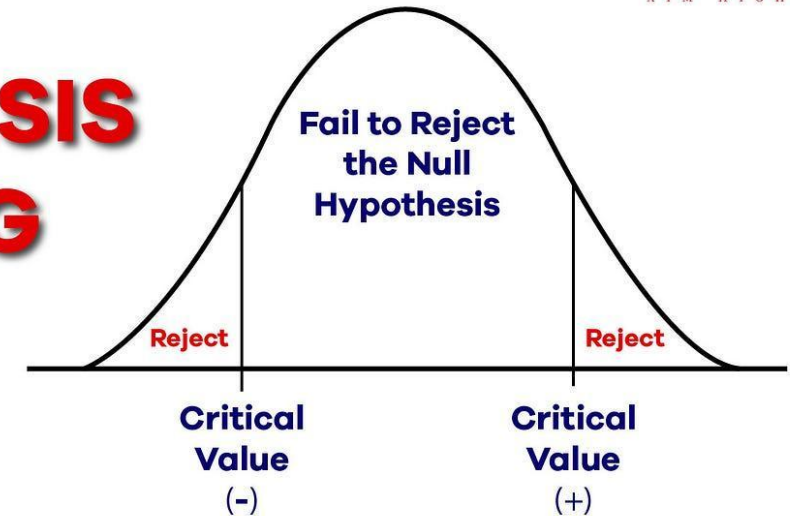


Hypothesis Testing

- P-Value
- Significance Level
- One Sample Z-Test
- Z-Statistic
- Null v Alternative Hypothesis



HYPOTHESIS TESTING



Can you think of some examples of when you have seen statistics/data analytics in your daily life?

Here are a few!

Advertisements: When you see an advertisement claiming that “4 out of 5 dentists recommend a particular toothpaste,” it is using statistics to persuade you. The advertisers conducted a survey or study and found that a majority of dentists (80%) recommend the toothpaste. This information is presented to make the product appear more reliable and effective.

News Reports: News outlets often use statistics to provide context and support their stories. For example, when reporting on unemployment rates, they might say that “unemployment has dropped by 2% over the last six months.” This statistic helps readers understand the scale and impact of the change in employment.

Digital Content: On social media platforms, statistics are often used to emphasize the importance of an issue or to generate interest. For instance, a post might say, “Did you know that on average, people spend 2 hours and 22 minutes on social media every day?” By presenting this information as a statistic, the post aims to grab your attention and potentially encourage a discussion about social media usage.



Many fields use statistics for different purposes, such as to help to keep us safe, improve our health, and advance our knowledge.

Example!

You were just selected as Mayor of a new town! Your town's population is 10,000 people. What ways will you apply statistics into developing your town? How can you apply data analytics to ensure your town's schools are large enough? How will you use data analytics to ensure your town runs successfully?